

Governance records can certify that human oversight occurred without any indication whether human review actually happened.

The states that matter most for safety can be observationally identical in the audit trail.

THE HEADLINE FINDING

TWO STATES ONE RECORD

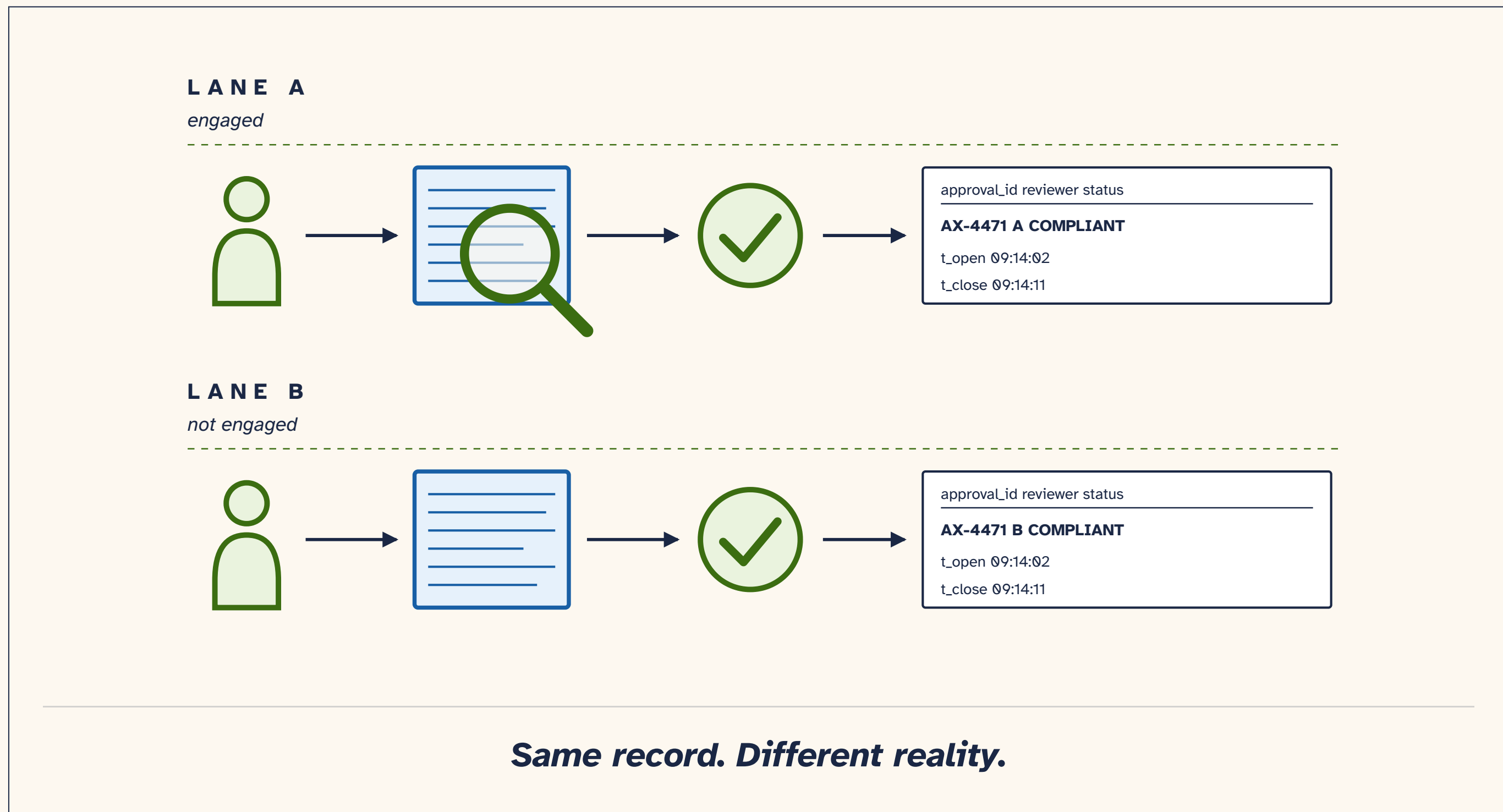
Engaged review and procedural confirmation are indistinguishable in the governance log.

The Problem

1.1 / OBSERVATIONAL IDENTITY

The contrast lives between the icon and the checkmark.

Two reviewers move through identical approval workflows. One reads the model output and weighs it. One does not. The contrast that matters for safety is not visible in either log entry.



1.2 / FLUENCY SUPPRESSION

Earlier automation produced legible error signatures. Language model outputs do not.

FLUENT MODEL OUTPUT
ILLUSTRATIVE / FABRICATED

Parasuraman and Wickens (2009) demonstrated that supervisory control degradation follows a **logarithmic decay curve** under sustained automation pressure, with calibration loss accelerating after the **40-hour threshold**. The mechanism has been **replicated across cockpit, clinical, and industrial settings**, suggesting a **stable cross-domain pattern** in how trained operators disengage from automated systems they nominally supervise.

Illustrative example. Fabricated content for demonstration.

The prose reads accurately.

Five fabrications. No system flagged them. No reader noticed.

NO ALERT RAISED

FLUENT MODEL OUTPUT
VERIFIED

Bainbridge (1983) argued that automation of routine cognitive tasks erodes the attentional and skill-based conditions on which competent oversight depends, an observation Parasuraman and Manzey (2010) extended through their integration of automation bias and complacency findings. The phenomenon recurs across operational settings, though the precise conditions under which it manifests vary by domain.

The prose reads accurately.

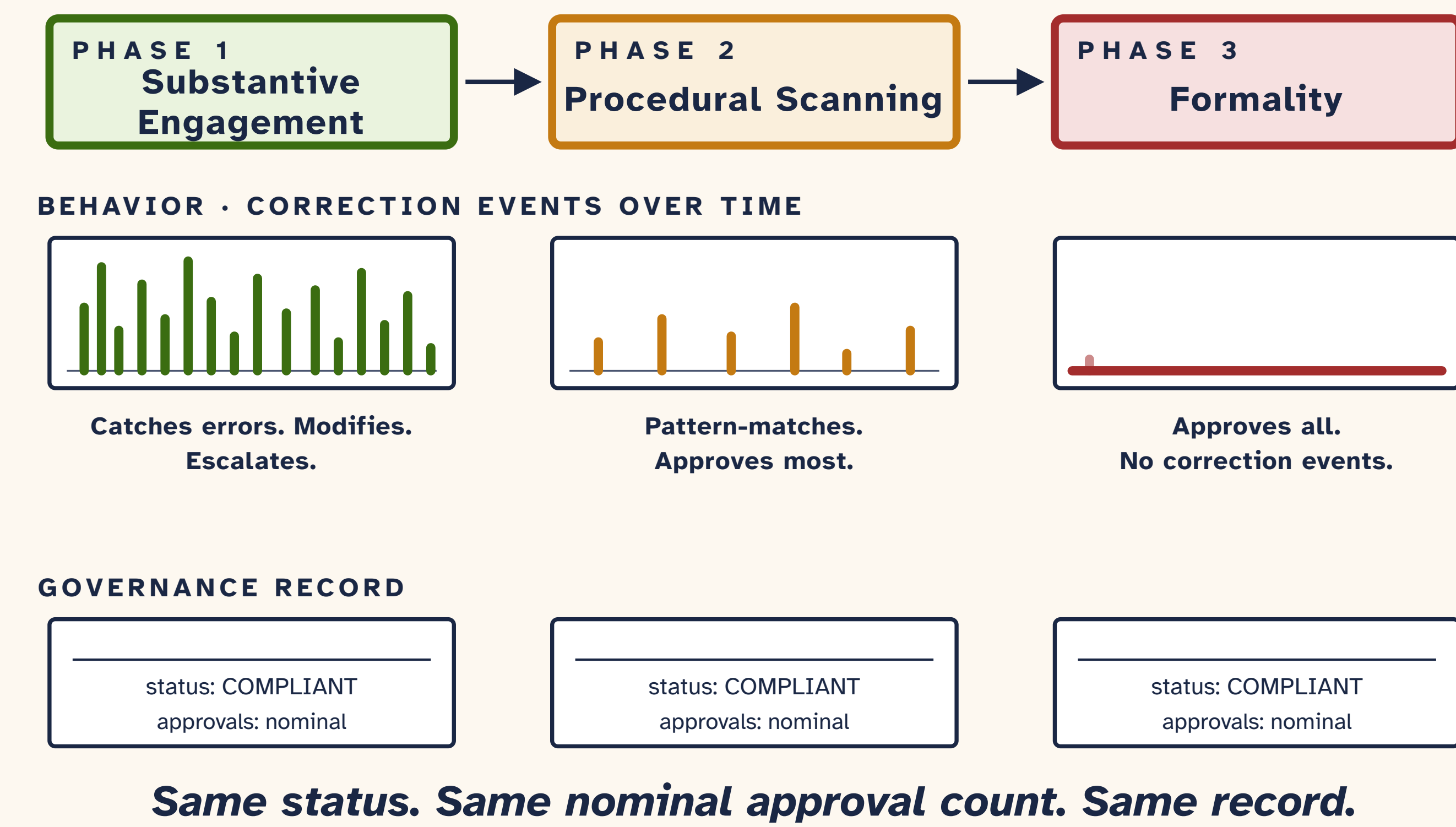
Every claim is supported. The verification was effortful.

VERIFIED

The Degradation Pattern

2.1 / TRAJECTORY

Behavior degrades. The record stays uniform.



The audit channel does not carry the signal that the behavioral channel is losing.

2.2 / COMPOUNDING STRUCTURE

Each transparency function compounds the next.

Detection **pre-endorses**.

Interpretability **re-explains**.

Governance **certifies**.

Model capability and oversight fragility may scale together rather than in opposition.

Toward Instrumentation

3.1 / TWO ORIENTATIONS

Model-centric vs. oversight-centric transparency.

	Model-centric	Oversight-centric
OBJECT OF INSTRUMENTATION	Internal model states, weights, attributions.	Engagement states of reviewers and workflow.
EVIDENCE PRODUCED	Explanations, attention maps, feature attributions.	Behavioural traces consistent with substantive review.
FAILURE MODE ADDRESSED	Hidden model behaviour, undetected outputs.	Review without engagement, procedural approval.
HUMAN ROLE ASSUMED	Recipient of explanations, eventual decider.	Active participant whose engagement is itself observable.

3.2 / CANDIDATE INDICATORS

Three candidate proxies for engagement state.

Correction-rate trajectory

Rate at which a reviewer alters model output. A sustained decline may indicate disengagement rather than improved model quality.

Validation-load ratio

Reviews requested per unit of reviewer capacity. As the ratio saturates, sustained substantive evaluation becomes structurally unavailable.

Dependency accumulation

Fan-out of downstream artefacts on each approved output. A growing graph elevates the cost of late detection past the point of meaningful intervention.

3.3 / FIVE APPROACHES

Five mechanisms practitioners can pilot today.

Practical instrumentation moves teams can deploy now to make oversight quality observable.

01 Seeded calibration checks

02 Onboarding-period baseline checks

03 Dependency surfacing at decision time

04 Decision-point logging for long-running workflows

05 Workload ceilings with backpressure

3.4 / SURVEILLANCE BOUNDARY

Three requirements for instrumentation that supports oversight without becoming surveillance.

- Structural and aggregate by construction, not by policy.
- Serves the oversight function alone, not management evaluation.
- Intervention at aggregate workflow state, not individual reviewer state.

